Comparative Analysis of BM25, Dense, and Hybrid Retrieval for Multilingual RAG Systems (English, French, Arabic)

Overview

This report evaluates three primary retrieval paradigms—BM25 (sparse lexical), dense vector retrieval, and hybrid retrieval—specifically within the context of a multilingual Retrieval-Augmented Generation (RAG) system handling English, French, and Arabic documents. Combining distinct linguistic typologies (Indo-European and Afroasiatic) introduces specific challenges such as Arabic's rich inflectional morphology, root-and-pattern structures, and script characteristics, contrasted against English and French. The evaluation focuses on retrieval accuracy, cross-lingual capabilities, and operational complexity.

Comparison

Criterion	Approach A	Approach B
Cross-Lingual and Multilingual Generalization	BM25 relies strictly on exact keyword matching per language, performing poorly on cross-lingual queries (e.g., querying in English to retrieve Arabic texts) without extensive translation pipelines.	Dense retrieval uses multilingual pre-trained language models (e.g., mDPR, multilingual sentence transformers) that map semantically equivalent concepts across English, French, and Arabic into a shared vector space, inherently supporting cross-lingual retrieval.
Handling of Arabic Morphological Complexity	Standard BM25 suffers significantly from Arabic's rich morphology unless paired with robust language-specific stemmers, light stemmers, or root-extractors, as un-stemmed word variations drop lexical overlap.	Dense retrieval captures semantic proximity robustly across Arabic word forms because contextual embedding models map inflected variants and synonyms to nearby latent coordinates.

Exact Keyword, Proper Noun, and Identifier Matching	BM25 excels at precision matching for exact terms, part numbers, acronyms, and rare proper nouns across all supported languages.	Dense retrieval often struggles with exact strings, acronyms, and rare codes due to subword tokenization limits and smoothing effects in continuous vector spaces.

Strengths & Weaknesses

Approach A

+ High precision for exact keyword matches, technical terminology, and proper names.
+ Requires zero GPU infrastructure and operates with predictable, low computational overhead.
+ Completely language-agnostic regarding training, functioning purely on inverted index frequencies.
– Fails completely on cross-lingual retrieval (searching in French to find Arabic documents) without translation layers.
– Highly sensitive to morphological variations in Arabic unless specialized stemmers are implemented.
– Blind to semantic synonyms and paraphrases within and across languages.

Approach B

+ Seamless cross-lingual and multilingual retrieval using shared embedding spaces.
+ Captures semantic synonyms, paraphrases, and conceptual alignment across English, French, and Arabic.
+ Robust to morphological variations and inflections without requiring custom language-specific stemmers.
– Prone to missing exact identifiers, acronyms, and rare proper nouns due to dense vector compression.
– Requires significant GPU memory and compute resources for embedding generation and Approximate Nearest Neighbor (ANN) search.
– Out-of-distribution performance can drop noticeably if the underlying encoder was not thoroughly fine-tuned on target multilingual datasets.

Recommendation

> For a multilingual RAG system incorporating English, French, and Arabic, a hybrid retrieval approach is strongly recommended. BM25 alone fails to bridge language barriers and struggles with Arabic's rich morphology, while dense retrieval alone often misses critical exact matches, serial numbers, and proper nouns. Utilizing a blended retriever (such as sparse lexical indices combined with multilingual dense embeddings) captures both the semantic and cross-lingual

> strengths of deep models and the exact-match precision of lexical search, satisfying the requirements of diverse linguistic typologies.

Sources

Mr. TyDi: A Multi-lingual Benchmark for Dense Retrieval
Demonstrates that while dense representations can lag behind BM25 in isolation for certain zero-shot tasks, sparse-dense hybrids significantly improve multi-lingual retrieval performance.
http://arxiv.org/abs/2108.08787v2

Synergistic Approach for Simultaneous Optimization of Monolingual, Cross-lingual, and Multilingual Information Retrieval
Provides training strategies and performance benchmarks for zero-shot multilingual retrieval across diverse language families.
http://arxiv.org/abs/2408.10536v1

Blended RAG: Improving RAG (Retriever-Augmented Generation) Accuracy with Semantic Search and Hybrid Query-Based Retrievers
Shows that combining dense vector indexes and sparse encoder indexes yields far superior downstream generative QA accuracy.
http://arxiv.org/abs/2404.07220v2